Canadian FHIR Interoperability Project Companion Guide

Use Cases, Standards Links, Data Examples, Delivery Assets, and Research References

Companion to the Canadian Interoperable FHIR Health Server SDLC

Purpose. This is a non-SDLC project companion. It consolidates the practical material needed to start and demonstrate a Canadian health-interoperability project: the problem framing, use cases, standard links, example FHIR payloads, terminology guidance, synthetic data scenarios, evaluation metrics, and a project launch checklist.

Important. This document is for technical planning, learning, prototyping, and approved pilot preparation. Do not use real personal health information in development or demonstrations without formal authorization, privacy controls, and appropriate agreements.

1. Project at a Glance

The project addresses the inability of authorized clinicians to reliably obtain structured, current, and clinically meaningful patient information when care crosses organizational or provincial boundaries. The objective is not to centralize every medical record. It is to create a secure, standards-based federated view that retrieves priority information, preserves source authority, shows provenance and freshness, and enforces consent-aware access.

Recommended MVP

Build a read-only inter-jurisdiction patient-summary retrieval workflow using synthetic data and two simulated gateways: one “Ontario” gateway and one “British Columbia” gateway. The system returns a validated patient summary with medication, allergy, condition, immunization, laboratory, and diagnostic-report data, including source and freshness information.

MVP element	Recommended choice
Clinical workflow	Emergency department retrieval of an external patient summary
Users	Authorized clinician, system administrator, privacy reviewer
Data	Synthetic or approved de-identified data only
Exchange	FHIR R4 Bundle/Composition using pinned PS-CA and CA Core+ baseline
Identity	Multiple identifiers plus human review of uncertain matches
Security	SMART on FHIR/OIDC, organization and purpose checks, audit, consent policy
Success	Retrieve a safe, interpretable, auditable summary with clear partial-response behavior

2. Canadian Standards Library

Pin exact versions in a standards-baseline file before implementation. Some Canadian implementation guides are published as draft or ballot versions; do not mix artifacts from multiple versions without documenting the decision.

Standard	Purpose	Primary link
HL7 FHIR R4	Base resource, REST API, search, Bundle, operation, history, and data-model standard	https://hl7.org/fhir/R4/index.html
CA Core+	Pan-Canadian FHIR profiles, extensions, and terminology artifacts	https://simplifier.net/guide/ca-core
CA Core+ technical context	Technical modules, artifacts, profile status, and implementation context	CA Core+ Technical Context
PS-CA	Pan-Canadian Patient Summary FHIR guidance	PS-CA FHIR Implementation Guide
PS-CA guide	PS-CA FHIR implementation-guide publication area and releases	https://simplifier.net/guide/ps-ca
CACDI	Canadian Core Data for Interoperability overview	CIHI CACDI overview
CACDI v2	Canadian core data content, elements, and value sets	CACDI Version 2 PDF
CACDI bulletin	Changes and context for CACDI v2	CACDI v2 bulletin

Standard	Purpose	Primary link
Canadian FHIR Registry	Canadian profiles, extensions, value sets, and reusable artifacts	Canadian FHIR Registry
Infoway FHIR community	Canadian FHIR collaboration, implementation activity, and working groups	Infoway FHIR Implementations

How the standards fit together

FHIR defines how data is represented and exchanged. CACDI defines a Canadian core set of data elements and value sets needed for connected care. CA Core+ supplies FHIR profiles, extensions, and terminology artifacts aligned to the pan-Canadian content framework. PS-CA applies this work to a structured patient-summary exchange. The project must use all four layers, not generic FHIR alone.

3. Core Clinical Use Cases

Use Case 1: Emergency patient-summary retrieval

Scenario. A patient who previously received care in British Columbia presents to an Ontario emergency department. The clinician requires current allergies, medications, active conditions, immunizations, recent labs, and diagnostic reports.

Step	System behaviour
1. Authenticate	Clinician authenticates through SMART on FHIR/OIDC; token is validated.
2. Authorize	System evaluates practitioner, organization, purpose of use, patient context, consent rules, and jurisdiction policy.
3. Identify	Identity service queries known identifiers, then evaluates matching candidates.
4. Retrieve	Ontario broker invokes a BC gateway adapter to request the permitted patient summary.
5. Validate	Returned FHIR Bundle is validated against the pinned profiles and terminology bindings.
6. Present	UI displays summary plus source organization, last update, freshness, and warnings.
7. Audit	System captures request, decision, response, source status, and correlation ID.

Failure modes to demonstrate: no patient match; ambiguous identity; consent denial; gateway timeout; invalid profile; stale cache; partial response; break-glass request; source-data conflict.

Use Case 2: Medication reconciliation during transition of care

Scenario. A patient is discharged from hospital and visits a community physician. Medication orders, patient-reported use, dispensing information, and allergies may conflict. The application retrieves source records, shows dates and provenance, and helps the clinician reconcile—not automatically overwrite—the medication list.

- Use MedicationRequest for orders/prescriptions, MedicationStatement for reported/current use, MedicationDispense for dispensing events, and MedicationAdministration for administered medications.
- Show duplicate or contradictory entries explicitly; do not collapse records without a clinician-approved reconciliation rule.
- Use a Canadian drug terminology strategy and preserve source medication codes alongside normalized codes.

Use Case 3: Laboratory and diagnostic-report sharing

Scenario. A specialist sees a patient who had tests performed in another jurisdiction. The specialist retrieves structured Observation and DiagnosticReport resources rather than relying on a faxed PDF.

- Use Observation for individual results, DiagnosticReport for the clinical report and its result references, and DocumentReference when a legacy document remains necessary.
- Preserve specimen, effective date, issuing laboratory, reference ranges, result status, and source terminology.

Use Case 4: Planned cross-provincial transition of care

Scenario. A patient moves from Ontario to British Columbia. The new care team needs a verified patient summary, care context, and source contacts. The system supports discovery and read-only summary retrieval while source systems remain authoritative.

Use Case 5: Privacy reviewer and break-glass audit

Scenario. A clinician uses emergency access because the patient cannot communicate. The system requires justification, limits the permitted scope and duration, creates an enhanced audit event, and routes the event for retrospective privacy review.

4. User Stories and Acceptance Criteria

User story	Acceptance criterion
As an emergency physician, I need to see external allergies before prescribing.	The summary visibly lists allergy source, status, source organization, and last update; unavailable source is shown as unavailable, not “no known allergies.”
As a clinician, I need to know if the patient match is reliable.	The system labels deterministic matches and candidate matches; uncertain matches cannot silently merge records.
As a privacy officer, I need to review cross-jurisdiction access.	Every protected access includes actor, organization, patient, purpose, decision, resources, source jurisdiction, timestamp, and correlation ID.
As a health information manager, I need to trace clinical facts to their source.	Each returned resource contains provenance and source-record references; transformations and terminology mappings are traceable.
As a patient, I need my information accessed only as allowed.	Policy checks consider consent and applicable restrictions; break-glass events are specially audited.
As an engineer, I need a clear failure response.	Invalid or unavailable remote data returns FHIR OperationOutcome plus source/partial-response indicators without exposing PHI in errors.

5. Recommended Data Domains

Domain	FHIR resources	MVP notes
Identity	Patient, Identifier, Organization	Multiple identifiers; record assigning authority and jurisdiction
Care team	Practitioner, PractitionerRole, Organization, CareTeam	Needed to establish source and request context
Conditions	Condition	Capture clinical and verification status; coded terminology
Allergies	AllergyIntolerance	Show verification, criticality, reaction, source, and freshness
Medications	MedicationRequest, MedicationStatement, MedicationDispense, MedicationAdministration	Do not use a single generic “medications” table
Immunizations	Immunization	Preserve product, date, status, source, and applicable terminology
Laboratory	Observation, DiagnosticReport	Use coded observation semantics and link report to results
Imaging/documents	DiagnosticReport, ImagingStudy, DocumentReference	Start with report metadata; guard Binary/document retrieval carefully
Patient summary	Composition, Bundle	Use the pinned PS-CA profiles and validation package
Trust	Consent, Provenance, AuditEvent, Endpoint, OperationOutcome	Required for safe and traceable federation

6. Synthetic Patient Scenarios

Use synthetic data for all early development and demos. Each test patient should represent a specific interoperability or safety condition—not just a happy-path record.

ID	Scenario	Expected system result
SYN-001	Exact provincial identifier match; complete summary	Retrieve and validate summary; record provenance and audit
SYN-002	Patient has two provincial identifiers after relocation	Link identifiers only under approved evidence; display source jurisdictions
SYN-003	Same name and date of birth as another patient	Return ambiguous candidate match; require human review; do not merge
SYN-004	Severe penicillin allergy in remote source	Display high-priority allergy with source and freshness
SYN-005	Medication order conflicts with patient-reported medication use	Show MedicationRequest and MedicationStatement separately for clinician reconciliation
SYN-006	Remote gateway is unavailable	Return explicit partial response; never imply no data exists
SYN-007	Summary has an invalid code or profile error	Reject/quarantine invalid item and return actionable OperationOutcome
SYN-008	Consent restriction applies	Deny or redact according to the policy; audit the decision
SYN-009	Emergency access requested	Require reason; apply expiry and enhanced audit/review
SYN-010	Source record is stale	Show source date, last refresh, and stale-data warning

7. FHIR Examples

7.1 Example Patient with provincial identifier

Illustrative only. Replace profile URLs and identifiers with the exact pinned implementation-guide artifacts selected for the project.

```
{ "resourceType": "Patient", "id": "synthetic-ontario-001", "meta": { "profile": [""], "lastUpdated":
"2026-08-25T12:00:00Z" }, "identifier": [{ "system": "https://example.ca/identifier/ontario-phn",
"value": "SYN-ON-000123", "use": "official" }], "name": [{"use": "official", "family": "Demo",
"given": ["Avery"]}], "birthDate": "1985-03-14", "gender": "other" }
```

7.2 Example AllergyIntolerance

```
{ "resourceType": "AllergyIntolerance", "id": "allergy-synthetic-001", "clinicalStatus": {"coding":
[{"code": "active"}]}, "verificationStatus": {"coding": [{"code": "confirmed"}]}, "type": "allergy",
"criticality": "high", "code": { "coding": [{ "system": "http://snomed.info/sct", "code": "91936005",
"display": "Allergy to penicillin" }] }, "patient": {"reference": "Patient/synthetic-ontario-001"},
"recordedDate": "2026-02-02", "note": [{"text": "Synthetic example only."}] }
```

7.3 Example Provenance

```
{ "resourceType": "Provenance", "target": [{"reference": "AllergyIntolerance/allergy-synthetic-001"}],
"recorded": "2026-08-25T12:05:00Z", "agent": [{ "type": {"text": "source organization"}, "who":
{"reference": "Organization/synthetic-bc-gateway"} }], "entity": [{ "role": "source", "what":
{"reference": "https://synthetic-bc.example.ca/fhir/AllergyIntolerance/remote-123"} }] }
```

7.4 Example partial-response OperationOutcome

```
{ "resourceType": "OperationOutcome", "issue": [{ "severity": "warning", "code": "transient",
"details": {"text": "British Columbia gateway did not respond within the configured timeout."},
"diagnostics": "Correlation ID: 9f8c1e2d" }] }
```

A production response should also return a governed source-status indicator so clinicians can distinguish “source unavailable” from “no relevant information found.”

8. Patient Summary Example Structure

At a high level, the summary should be a FHIR document Bundle whose first entry is a Composition and whose remaining entries include the patient and the clinical resources referenced by the Composition. Implement the exact PS-CA constraints from the version you select.

```
Bundle (type: document) ■■■ Composition (patient summary document) ■■■ Patient ■■■
AllergyIntolerance (0..*) ■■■ MedicationStatement / MedicationRequest (0..*) ■■■ Condition (0..*)
■■■ Immunization (0..*) ■■■ Observation / DiagnosticReport (as permitted by scope) ■■■ Practitioner
/ Organization (as required for attribution) ■■■ Provenance (source and transformation information)
```

Clinical display rules

- Show the source organization and source system for each high-risk clinical fact.
- Show effective clinical time separately from the last synchronization time.
- Label data as live, cached, imported, partial, stale, unverified, or conflicting as applicable.
- Never convert a gateway failure into a “no known allergies” or “no current medications” assertion.
- Preserve source terminology and record normalized terminology separately.

9. Identity and Consent Examples

9.1 Identity matching decision table

Match level	Example	Action
Deterministic	Trusted identifier returned by authoritative source	May retrieve according to policy
Strong demographic	Name, date of birth, and multiple verified identifiers agree	Return candidate; apply local policy before link
Probabilistic	Similar name, DOB, historical address; score 0.82	Human review required; no automatic merge
Low confidence	Only name or name + DOB; conflicting demographic data	Do not link or retrieve as matched patient

9.2 Authorization decision context


```
{ "requester": { "practitioner": "Practitioner/ed-physician-001", "organization":
"Organization/ontario-emergency-department", "role": "emergency-physician" }, "patient":
"Patient/synthetic-ontario-001", "purposeOfUse": "TREATMENT", "requestedResources":
["AllergyIntolerance", "MedicationStatement", "Condition"], "sourceJurisdiction": "BC",
"requestingJurisdiction": "ON", "emergencyOverride": false, "correlationId": "9f8c1e2d" }
```


The policy service should produce an explicit permit, deny, redact, or break-glass decision plus obligations such as required audit fields, resource restrictions, or expiry time.

10. Architecture Reference

Use this as a project discussion diagram, not as a final production blueprint.

```
Clinician EMR / Portal / Authorized App | API Gateway + Rate Limits | SMART on FHIR / OIDC
Authentication | Consent + Policy Decision Service | Patient Identity Resolution | Federation Broker /
Router / | \ Ontario Adapter BC Adapter Test/Sandbox Adapter \ | / Terminology Service | FHIR Store +
Search Projections | Provenance + Audit + Monitoring | PostgreSQL / Secure Object Store / Bounded Redis
Cache
```

Component responsibilities

Component	Responsibility
FHIR server	Stores/serves validated FHIR resources, searches, operations, resource history
Policy engine	Evaluates organization, role, purpose, patient context, consent, resource, and jurisdiction rules
Identity service	Stores identifiers, generates candidate matches, supports review and merge/unmerge governance
Gateway adapters	Encapsulate jurisdiction-specific endpoint, auth, profile, request, and error differences
Terminology service	Validates codes, expands value sets, translates/localizes concepts, manages versions
Provenance/audit service	Preserves clinical source lineage and security/access events
Cache	Caches only policy-approved data; never becomes a separate authorization authority
Observability	Tracks latency, gateway health, errors, profile failures, identity uncertainty, and security events

11. Project Deliverables Checklist

This companion does not replace your SDLC. Use the following as the project artifact checklist outside the core SDLC.

Area	Key deliverables
Clinical	Use-case specification; workflow maps; clinical safety hazards; usability report; training scenarios
Standards	Standards baseline; FHIR implementation guide; PS-CA summary spec; CA Core+ mapping; CACDI matrix; terminology strategy
Governance	Privacy impact assessment; threat/risk assessment; data flow; consent policy; data-sharing model; Indigenous governance engagement plan
Architecture	Target architecture; identity design; gateway contract; database model; caching policy; audit/provenance design
Engineering	CapabilityStatement; API contract; sample Bundles; source code; infrastructure as code; CI/CD pipeline
Testing	Synthetic data; FHIR validation; terminology suite; identity safety suite; security suite; gateway contracts; performance/resilience report
Operations	Monitoring plan; incident runbooks; backup/DR plan; on-call/support model; release/change process
Pilot	Pilot plan; go-live checklist; benefits dashboard; post-pilot evaluation; scale-out plan

12. Evaluation Metrics

Metric area	Example measure	Why it matters
Data availability	% of eligible summaries retrieved successfully	Measures whether information moves when needed
Completeness	% of required summary elements present and valid	Prevents a technically successful but clinically empty result
Freshness	Median age of medication and allergy data	Makes stale-data risk visible
Identity safety	False-match, missed-match, and human-review rates	Protects against wrong-patient disclosure and clinical error
Workflow	Time to retrieve external summary; manual re-entry/fax reduction	Measures real clinician benefit
Safety	Medication discrepancies, unavailable allergy data, partial-response acknowledgement	Measures patient-safety outcomes
Privacy	Audit completeness; policy denials; break-glass review completion	Demonstrates accountable access
Reliability	Gateway success rate, p95 latency, outage recovery time	Measures federation readiness
Adoption	Active users, organizations, and recurring use	Tests whether the solution fits care delivery

13. Learning and Implementation Resources

Topic	Recommended resource
FHIR basics	HL7 FHIR R4 specification: hl7.org/fhir/R4
FHIR resource examples	FHIR R4 examples and resource pages within the official specification
Canadian FHIR profiles	Canadian FHIR Registry: simplifier.net/organization/canadianfhirregistry
CA Core+	CA Core+ IG: simplifier.net/guide/ca-core
Patient summary	PS-CA IG overview: Infoway PS-CA guide
Canadian data content	CIHI CACDI overview: CIHI CACDI
Canadian Connected Care	CIHI Connected Care: cihi.ca/en/connected-care
Canadian collaboration	Infoway FHIR implementations: InfoCentral FHIR group

14. First 30-Day Plan

Week	Outcome
Week 1	Choose one MVP use case; create charter, standards baseline, stakeholder map, and synthetic patient scenarios.
Week 2	Draft FHIR resource map, PS-CA summary structure, CACDI-to-FHIR matrix, terminology strategy, and identity rules.
Week 3	Create architecture, gateway adapter interface, consent/policy model, provenance/audit model, and test strategy.
Week 4	Build local FHIR server with synthetic Patient, AllergyIntolerance, Condition, MedicationStatement, Observation, DiagnosticReport, Bundle, Provenance, and AuditEvent; validate sample summary.

Definition of a successful first demo

An authorized simulated Ontario clinician retrieves a synthetic patient summary from a simulated BC gateway. The response passes the pinned profile and terminology validation rules; shows allergies, medications, conditions, and recent results; displays source and freshness; distinguishes a partial response from “no data”; avoids automatic matching of ambiguous patients; and produces a complete audit trail.

15. Reference Notes

Primary project references include the official HL7 FHIR R4 specification, Canada Health Infoway’s CA Core+ and PS-CA implementation materials, the Canadian FHIR Registry, and CIHI’s CACDI/Connected Care publications. CACDI defines a standardized set of essential health data elements and value sets, while CA Core+ provides foundational FHIR profiles, extensions, and terminology artifacts aligned to the pan-Canadian health-data-content work.

Standards and implementation guides evolve. Verify publication status, licensing, package names, and version compatibility with the relevant Canadian authority or project partner before production implementation. This guide intentionally does not provide legal advice or assert that use of a Canadian hosting region alone establishes privacy-law compliance.

End of companion guide.